

Problem–Solution Fit

Project: Predictive Risk Clustering and Visual Informatics for Cardiovascular Disease (HeartInsight)

Problem Statement

Healthcare professionals often work with large clinical datasets that are difficult to interpret quickly. Traditional reports hide relationships among age, BMI, diabetes, stroke history, and lifestyle factors, delaying clinical decisions.

Proposed Solution

HeartInsight uses Tableau Public and Flask to transform the Heart_new2.csv dataset (4,494 patient records, 18 clinical features) into interactive dashboards for demographics, lifestyle, co-morbidities, and metabolic risk analysis.

Problem–Solution Fit Matrix

Problem: Static reports are difficult to analyze.

Solution: Interactive dashboards with filtering and visualization.

Value: Faster insights, improved risk identification, and better decision support.

Benefits

- Quick visualization of patient trends.
- Interactive filtering by age, gender, BMI, diabetes, smoking, alcohol, and stroke.
- Easy browser-based access.
- Supports clinicians and health analysts.

Conclusion

HeartInsight converts complex cardiovascular data into meaningful visual insights, helping healthcare professionals make faster and more informed decisions.